AI-Powered Due Diligence & Compliance Risk Dashboard

Phase-Wise Development Plan

Live React analytics dashboard with Python data pipeline and optional RAG assistant

Prepared as a portfolio-ready execution plan for an MSBA-oriented compliance analytics project.

Document focus: data mining, explainable risk scoring, analytics dashboard, client 360 view, deployment, and AI/RAG extension.

Project Area	Description
Business Domain	KYC, AML, customer due diligence, enhanced due diligence, compliance risk management
Primary Outcome	A live dashboard that turns fragmented due diligence data into a unified client risk view
Recommended Stack	React or Next.js, TypeScript, Tailwind CSS, Recharts, Python, Pandas, RapidFuzz, Vercel
Development Method	Phase-wise MVP first, then reports and RAG as advanced releases

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1. Project Development Strategy

This development plan converts the technical idea into a practical phase-wise build. The recommended approach is to create a working live dashboard first, then gradually add advanced features such as PDF report generation and RAG.

Recommended Build Order

```
Data Foundation
  -&gt; Risk Scoring Logic
  -&gt; React Dashboard
  -&gt; Client 360 View
  -&gt; Report Generation
  -&gt; Live Deployment
  -&gt; RAG Assistant
```

The project should always have a working version at the end of each major phase. This reduces risk and helps you show progress early in a portfolio, interview, or MSBA application discussion.

Core principle: do not start with RAG first. The dashboard, data model, and explainable risk scoring should be completed before adding the AI assistant.

2. Final Target Product

The final product will be a live web dashboard where users can view compliance risk analytics, prioritize cases, review client-level due diligence evidence, generate a report, and optionally ask an AI/RAG assistant questions about a client.

Capability	User Value
Compliance KPIs	Managers can quickly understand alert volume, high-risk clients, open cases, and risk distribution.
Case Queue	Analysts can prioritize high-risk and aging cases instead of manually searching through files.
Client 360 View	All relevant due diligence data is visible in one place for a selected client.
Risk Score Breakdown	Analysts can understand why a client is classified as Low, Medium, High, or Critical risk.
Due Diligence Report	The system generates a standardized investigation summary to reduce manual reporting effort.
RAG Assistant	Analysts can ask natural-language questions and receive evidence-backed responses.

3. Recommended Tech Stack

Frontend

Area	Recommended Tool
Framework	Next.js or React Vite
Language	TypeScript preferred
Styling	Tailwind CSS
Charts	Recharts
Tables	TanStack Table or custom reusable table
Deployment	Vercel

Data Processing

Area	Recommended Tool
Data generation	Python
Data cleaning	Pandas
Entity matching	RapidFuzz
Risk scoring	Python rule-based logic
Output format	JSON files consumed by React

Optional AI/RAG

Area	Recommended Tool
Backend	FastAPI or Next.js API routes
Vector database	ChromaDB or FAISS
LLM	OpenAI API or local model
Retrieval framework	LangChain or LlamaIndex

4. Phase 0 - Project Planning and Scope Lock

Objective

Define exactly what the first live version will include so the project does not become too large. This phase prevents scope creep and gives the project a clear MVP.

Recommended MVP Scope

```
Executive Dashboard
Case Queue
Client 360 View
Risk Analytics
Due Diligence Report
Live Deployment
```

RAG should be treated as a later advanced feature unless you already have the core dashboard and data pipeline working.

Tasks

- Finalize project name and positioning statement.
- Finalize whether the frontend will use Next.js or React Vite.
- Finalize whether code will be TypeScript or JavaScript.
- Finalize dashboard pages for MVP.
- Finalize dataset size and dataset schemas.
- Finalize risk factors and risk scoring bands.
- Finalize dashboard KPIs and core charts.
- Create GitHub repository and initial README.
- Create top-level folder structure for data, scripts, source code, and reports.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Project scope	MVP and post-MVP features are clearly separated.
GitHub repo	Repository exists with initial folder structure.
README draft	Problem statement, proposed solution, and architecture are described.
Development checklist	A clear list of phases and tasks is ready.

5. Phase 1 - Data Model and Synthetic Dataset Creation

Objective

Create realistic synthetic compliance datasets that represent the due diligence workflow. This is the foundation for data mining, analytics, dashboarding, and RAG. Real bank or client data should not be used.

Datasets to Create

```
clients.csv
ownership.csv
screening_hits.csv
country_risk.csv
adverse_media.csv
previous_cases.csv
```

Recommended Dataset Size

Dataset	Record Count for MVP
Clients	100 to 300
Ownership / Board Members	300 to 800
Screening Hits	150 to 500
Adverse Media Articles	50 to 200
Previous Cases	50 to 150
Country Risk Records	50 to 100

Dataset 1: clients.csv

Field	Description
client_id	Unique client identifier
client_name	Legal name of the client
client_type	Corporate, Individual, Trust, Fund, or other type
industry	Business industry or sector
incorporation_country	Country of incorporation or registration
operating_countries	Countries where the client operates, separated by semicolon
client_status	Active, Offboarded, Quarantined, Restricted
relationship_status	Existing, New, Dormant
annual_revenue	Approximate annual revenue
onboarding_date	Date the client was onboarded
existing_risk_rating	Existing risk rating, if available

```
Example row:
C001, ABC Global Holdings, Corporate, Logistics, UAE, UAE;Russia;Singapore, Active, Existing, 12000000, 2021-
Medium
```

Dataset 2: ownership.csv

```
entity_id
client_id
entity_name
role
ownership_percentage
nationality
residence_country
pep_flag
sanction_flag
adverse_media_flag
```

Dataset 3: screening_hits.csv

```
hit_id
client_id
matched_entity_name
internal_entity_name
hit_type
hit_country
severity
hit_date
source_reference
initial_status
```

Allowed Hit Type	Description
Sanction	Potential sanctions list exposure
PEP	Politically exposed person association
Adverse Media	Negative news or media-related risk
Internal Alert	Internal watchlist or bank-specific alert
Regulatory	Regulatory or enforcement-related alert

Dataset 4: country_risk.csv

```
country
country_risk_level
sanctions_exposure
fatf_style_risk
corruption_risk
conflict_zone_flag
regulatory_risk_flag
```

Dataset 5: adverse_media.csv

```
article_id
client_id
entity_mentioned
article_title
article_text
source
publication_date
risk_category
severity
sentiment
```

Risk Category	Examples
Fraud	Financial misrepresentation, fake invoices, fraudulent activity
Bribery	Bribe allegations or investigations
Corruption	Public corruption or misuse of influence
Money Laundering	Alleged laundering activity or facilitation
Sanctions Evasion	Avoiding sanctions or restricted trade activity
Regulatory Violation	Regulatory warning, enforcement, or penalty
Tax Evasion	Tax fraud or tax evasion allegations
No Material Risk	Article does not materially affect risk assessment

Dataset 6: previous_cases.csv

```
case_id
client_id
previous_hit_type
previous_decision
analyst_notes
final_action
closure_date
reviewer_comments
```

Synthetic Data Design Tasks

- Create realistic synthetic records across all six datasets.
- Include low-risk, medium-risk, high-risk, and critical-risk client scenarios.
- Add intentionally messy data: missing countries, duplicate names, similar names, conflicting statuses, inconsistent spelling, and missing source references.
- Create at least one critical sanctions case, one high adverse media case, one PEP case, one false positive case, and one complex ownership case.
- Document every dataset in a data dictionary so the project is understandable to non-technical reviewers.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Raw CSV files	All six CSV files are created and stored under data/raw.
Data dictionary	Every field has a clear definition.
Sample client scenarios	At least five strong due diligence cases are documented.
Variation for charts	Data has enough variation for risk, country, industry, and alert distribution analytics.

6. Phase 2 - Python Data Processing Pipeline

Objective

Build a Python script that cleans raw CSV files and creates processed JSON files for the React dashboard. This phase demonstrates the end-to-end data mining and data preparation layer.

Pipeline Flow

```
Load raw CSV files
  -&gt; Validate required columns
  -&gt; Clean text fields
  -&gt; Standardize countries
  -&gt; Normalize dates
  -&gt; Handle missing values
  -&gt; Remove duplicates
  -&gt; Merge data
  -&gt; Create client-level features
  -&gt; Export processed JSON
```

Data Loading Requirements

The script should read all raw CSV files using Pandas and validate that they exist before processing. Missing files should produce clear error messages.

```
clients.csv
ownership.csv
screening_hits.csv
country_risk.csv
adverse_media.csv
previous_cases.csv
```

Validation Rules

Rule	Purpose
client_id must exist in all client-linked files	Ensures datasets can be merged.
client_name cannot be empty	Ensures client profiles are usable.
hit_type must be from allowed values	Prevents inconsistent alert classification.
severity must be Low, Medium, High, or Critical	Ensures risk scoring can run consistently.
ownership_percentage cannot exceed 100	Flags ownership data issues.
dates must be valid and standardized	Enables trend and aging analysis.

Cleaning Logic

- Trim extra spaces from all text fields.
- Convert names to consistent case for matching.
- Standardize country names.

- Normalize company suffixes such as Ltd, Limited, LLC, GmbH, Pvt Ltd.
- Convert dates to YYYY-MM-DD format.
- Remove duplicate client, ownership, and hit records.
- Fill missing values where reasonable.
- Flag missing critical values rather than hiding them.

Derived Client-Level Features

```
total_hits
sanction_hits
pep_hits
adverse_media_hits
internal_alert_hits
regulatory_hits
high_severity_hits
critical_severity_hits
high_risk_country_count
critical_risk_country_count
board_member_count
ubo_count
pep_associated_entities
sanction_associated_entities
adverse_media_article_count
previous_true_hits
previous_false_positives
```

Processed JSON Outputs

Output File	Purpose
dashboard_summary.json	Stores top-level KPIs and summary metrics.
client_risk_master.json	Stores one row per client with risk features and score.
case_queue.json	Stores case-level records for the analyst queue.
client_profiles.json	Stores detailed client 360 information.
risk_analytics.json	Stores chart-ready aggregations for analytics pages.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
process_data.py	Script runs end-to-end without errors.
Cleaned output	Cleaned CSV or intermediate output is produced.
Processed JSON	React-ready JSON files are generated.
Data quality report	Missing values, duplicates, and validation issues are summarized.

7. Phase 3 - Entity Matching and False Positive Detection

Objective

Add fuzzy matching logic to compare screening names with internal client or entity names. This is a realistic AML/KYC feature because many alerts are false positives caused by similar names.

Matching Inputs

```
matched_entity_name
internal_entity_name
hit_country
client_country
date_of_birth, if available
registration number, if available
nationality, if available
role in company
```

Core Matching Logic

Use RapidFuzz to calculate similarity between the screening name and the internal name. The output should be a numeric score plus a business-friendly classification.

```
Screening Name: Mohammad Ali Khan
Internal Name: Mohammed A. Khan
Similarity Score: 86
Classification: Potential True Match
```

Similarity Score	Classification
0 to 60	Likely False Positive
61 to 85	Possible Match
86 to 100	Potential True Match

Output Fields

```
hit_id
client_id
matched_entity_name
internal_entity_name
similarity_score
country_match
match_classification
match_explanation
```

Tasks

- Create entity_matching.py.
- Calculate similarity score for every screening hit.
- Classify every hit into likely false positive, possible match, or potential true match.
- Add country match logic when country fields are available.
- Add a plain-English explanation for each match classification.
- Add matching fields to case queue and client profile JSON outputs.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
entity_matching.py	Every screening hit has a similarity score.
Match classification	Every hit has a clear classification.
False positive field	Dashboard can filter likely false positives.
Match explanation	Client 360 page can explain why a match is possible or unlikely.

8. Phase 4 - Risk Scoring Engine

Objective

Create an explainable risk score for every client. The score should be easy to understand, auditable, and suitable for dashboard storytelling.

Risk Scoring Model

Risk Factor	Score
Sanctions hit	+100
PEP association	+60
High adverse media	+50
Medium adverse media	+30
Critical country exposure	+50
High-risk country exposure	+30
Complex ownership structure	+25
Internal alert	+20
Previous true hit	+40
Previous false positive	-10
Active relationship	+10
Offboarded client	-20

Risk Level Classification

Risk Score	Risk Level
0 to 40	Low
41 to 70	Medium
71 to 100	High
100+	Critical

Recommendation Logic

Risk Level	Recommendation
Low	Continue Relationship
Medium	Monitor
High	Enhanced Due Diligence
Critical	Escalate / Restrict / Offboard

Required Client Output

```
risk_score
risk_level
recommended_action
risk_score_breakdown
top_risk_drivers
```

Example Risk Explanation

```
Client: ABC Global Holdings
Risk Score: 110
Risk Level: Critical

Breakdown:
+50 High-severity adverse media
+30 High-risk country exposure
+20 Internal alert
+10 Active relationship

Recommendation:
Escalate for senior compliance review
```

Tasks

- Create risk_scoring.py.
- Implement rule-based scoring logic.
- Add risk score to the client master dataset.
- Add risk level and recommended action.
- Create detailed risk score breakdown for every client.
- Identify top three risk drivers for each client.
- Export updated JSON files for dashboard use.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
risk_scoring.py	Risk score is calculated for every client.
Risk levels	Each client has Low, Medium, High, or Critical classification.
Recommendation	Each client has a recommended action.
Score breakdown	Client 360 page can explain all score contributions.

9. Phase 5 - React Application Setup

Objective

Set up the live dashboard application structure and ensure the React app can load processed JSON data from the public folder.

Recommended Folder Structure

```
due-diligence-risk-dashboard/
  public/
    data/
      dashboard_summary.json
      client_risk_master.json
      case_queue.json
      client_profiles.json
      risk_analytics.json
  src/
    app/
      page.tsx
      cases/
      clients/
      analytics/
      reports/
      rag/
    components/
      KpiCard.tsx
      Sidebar.tsx
      Header.tsx
      RiskBadge.tsx
      RiskDistributionChart.tsx
      CaseQueueTable.tsx
      ClientProfileCard.tsx
      RiskScoreBreakdown.tsx
      ReportPreview.tsx
    lib/
      data.ts
      utils.ts
      constants.ts
    types/
      index.ts
  scripts/
    process_data.py
    risk_scoring.py
    entity_matching.py
  data/
    raw/
    processed/
  README.md
  package.json
```

Pages to Create

Route	Purpose
/	Dashboard homepage / executive overview
/cases	Analyst case queue
/clients	Client list
/clients/[clientId]	Client 360 view
/analytics	Risk analytics
/reports	Due diligence report generation
/rag	Optional RAG assistant page

Tasks

- Create Next.js or React app.
- Install Tailwind CSS.
- Install charting library such as Recharts.
- Create global layout and navigation.

- Create sidebar navigation.
- Create reusable KPI card component.
- Create reusable risk badge component.
- Create data loading utility to fetch JSON from public/data.
- Create empty page shells for all MVP routes.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Running React app	App runs locally without errors.
Navigation layout	All routes are accessible from sidebar or header.
Data loading utility	JSON files can be loaded in the browser.
Reusable components	KPI card, risk badge, and basic table components exist.

10. Phase 6 - Executive Dashboard Page

Objective

Build the main dashboard page for management-level analytics. This should immediately communicate business value and risk visibility.

Page URL

```
/
```

KPI Cards

```
Total Clients
Total Screening Hits
High-Risk Clients
Critical-Risk Clients
Open Cases
False Positive Rate
True Hit Rate
Average Risk Score
```

Charts

```
Risk Level Distribution
Hit Type Distribution
Country Risk Exposure
Monthly Alert Trend
Client Status Breakdown
```

Top 10 High-Risk Clients Table

Column	Description
Client Name	Name of client
Industry	Client sector
Country	Incorporation or primary operating country
Risk Score	Calculated score
Risk Level	Low, Medium, High, Critical
Recommended Action	System recommendation based on risk level

Optional Filters

```
Country
Risk Level
Industry
Client Status
Date Range
```

Tasks

- Create KPI card layout.
- Build risk distribution chart.
- Build hit type distribution chart.
- Build country exposure chart.
- Build monthly alert trend chart.
- Build client status breakdown chart.
- Build top high-risk clients table.
- Make the page responsive for laptop and tablet screens.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Executive dashboard	Dashboard gives a clear overview of compliance risk.
KPI cards	Metrics are calculated correctly from JSON data.
Charts	Charts render correctly and use real processed data.
Top-risk table	High-risk clients are easy to identify.

11. Phase 7 - Case Queue Page

Objective

Create an analyst work queue that helps prioritize cases by risk score, case age, hit type, and match classification.

Page URL

```
/cases
```

Table Columns

```
Case ID
Client Name
Hit Type
Risk Score
Risk Level
Match Classification
Case Status
Country
Case Age
Recommended Action
```

Filters

```
Risk Level
Hit Type
Case Status
Country
Client Status
Match Classification
```

Sorting

```
Risk Score
Case Age
Hit Date
Risk Level
```

Row Behavior

When the user clicks a case row, the app should open the client profile or a case detail drawer. For MVP, navigation to the Client 360 page is sufficient.

Tasks

- Build case queue table.
- Add filters for risk level, hit type, case status, country, client status, and match classification.
- Add search by client name.
- Add risk level badges.
- Add match classification badges.
- Add recommended action column.
- Add row click navigation to client profile.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Case queue page	Cases are displayed in a risk-prioritized table.
Filters and search	Users can narrow cases by relevant fields.
Sorting	Users can sort by risk score and case age.
Navigation	Users can open a client profile from the case row.

12. Phase 8 - Client 360 Page

Objective

Create a detailed client-level profile page showing all due diligence evidence in one place. This is the strongest page for demonstrating domain knowledge.

Page URL

```
/clients/[clientId]
```

Client Summary Section

```
Client Name
Client ID
Client Type
Industry
Incorporation Country
Operating Countries
Client Status
Relationship Status
Annual Revenue
Onboarding Date
Existing Risk Rating
```

Risk Score Summary

```
Risk Score
Risk Level
Recommended Action
Top Risk Drivers
```

Risk Score Breakdown

Field	Description
Risk Factor	The specific indicator contributing to risk
Score Contribution	Numeric impact on total score
Explanation	Plain-English reason for the contribution

```
Example:
High adverse media: +50
High-risk country exposure: +30
Internal alert: +20
```

Ownership and Board Members

```
Entity Name
Role
Ownership Percentage
Nationality
Residence Country
PEP Flag
Sanction Flag
Adverse Media Flag
```

Screening Hits

```
Hit Type
Matched Name
Internal Name
Similarity Score
Match Classification
Severity
Hit Country
Hit Date
Source Reference
```

Adverse Media

```
Article Title
Entity Mentioned
Risk Category
Severity
Sentiment
Publication Date
Article Summary
```

Previous Cases

```
Previous Hit Type
Previous Decision
Final Action
Closure Date
Analyst Notes
Reviewer Comments
```

Tasks

- Create dynamic client route.
- Load selected client profile from client_profiles.json.
- Build client summary card.
- Build risk score card.
- Build risk score breakdown component.
- Build ownership table.
- Build screening hits table.
- Build adverse media section.
- Build previous cases section.
- Add button to generate due diligence report for selected client.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Client 360 page	User can open any client and view complete due diligence information.
Risk score explanation	User can understand why the client is risky.
Evidence sections	Ownership, hits, adverse media, and previous cases are visible.
Report link	User can move from client profile to report generation.

13. Phase 9 - Risk Analytics Page

Objective

Create deeper analytics views for MSBA-style insights and management-level decision-making. This page should answer business questions, not just display charts.

Page URL

```
/analytics
```

Analytics to Show

```
Risk by Industry
Risk by Country
False Positive Rate by Hit Type
True Hit Rate by Hit Type
Average Risk Score by Country
Top High-Risk Jurisdictions
PEP vs Sanctions vs Adverse Media Distribution
Client Status vs Risk Level
Case Aging Distribution
```

Suggested Charts

Insight	Chart Type
Risk by industry	Bar chart
Risk by country	Bar chart or map
Alert distribution	Pie or bar chart
Monthly trend	Line chart
Case aging	Histogram or bar chart
False positives by hit type	Bar chart

Business Interpretation

Each chart should include a short insight sentence below it. This makes the dashboard stronger for MSBA because it communicates decision-making, not just visualization.

> Example insight: Adverse media alerts contribute to the largest share of high-risk cases, while PEP alerts show a higher false-positive rate. This suggests analyst effort can be optimized by prioritizing adverse media and sanctions cases first.

Tasks

- Build analytics data transformations in Python or frontend utility functions.
- Create risk by industry chart.
- Create risk by country chart.
- Create false positive rate by hit type chart.
- Create case aging distribution chart.
- Create high-risk jurisdiction chart.
- Add filters for country, industry, risk level, and date range.
- Add short business interpretation under each chart.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Risk analytics page	Page answers meaningful business questions.
Charts	Charts are clear and interpretable.
Insight text	Charts are connected to compliance decisions.
Filters	Users can explore segments of risk data.

14. Phase 10 - Due Diligence Report Page

Objective

Allow users to generate a standardized due diligence report for a selected client. This connects the dashboard directly to the analyst workflow.

Page URL

```
/reports
```

Report Sections

```
Client Summary
Business Activity
Ownership Review
Screening Hit Details
Entity Matching Result
Adverse Media Review
Country Risk Exposure
Previous Case History
Risk Score Explanation
Final Recommendation
Analyst Notes
```

Report Generation Logic

The report should use client profile data and generate structured narrative text. For MVP, a deterministic template is better than AI-generated text because it is predictable and explainable.

```
ABC Global Holdings is an active corporate client operating in the logistics sector with exposure to UAE, Rus
Singapore. The system identified one high-severity adverse media article related to bribery and one internal
a previous review cycle.

The client received a risk score of 92 and is classified as High Risk. The main risk drivers are adverse medi
 high-risk jurisdiction association, and previous internal alert history.

Recommendation: Enhanced Due Diligence and senior compliance review before continuing the relationship.
```

MVP Report Options

Option	Use in MVP?
Show report preview on screen	Yes
Copy to clipboard	Yes
Download as text or HTML	Yes
Download as PDF	Can be added after MVP
Analyst notes field	Recommended
Reviewer comments field	Optional

Tasks

- Create report page.
- Add client selector.
- Generate report text from selected client data.
- Add risk score breakdown to report.
- Add evidence references.
- Add copy or download button.
- Add analyst notes field.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Report page	User can select a client.
Report preview	Complete due diligence report is generated.
Risk explanation	Report includes score breakdown and top risk drivers.
Recommendation	Report includes final recommended action.
Export option	Report can be copied or downloaded.

15. Phase 11 - Live Deployment

Objective

Deploy the React dashboard and make it accessible through a public URL. This is essential for portfolio value because reviewers can interact with the project directly.

Recommended Deployment Platform

Use Vercel for the React or Next.js application. It integrates well with GitHub and is simple for static JSON-based dashboards.

Deployment Steps

- Push code to GitHub.
- Connect GitHub repository to Vercel.
- Configure build command.
- Configure output directory if required.
- Deploy the app.
- Test all pages on the live URL.
- Fix routing or static data path issues.
- Add live URL to README.

Pre-Deployment Checklist

```
App builds successfully.
No console errors.
All JSON files are available in public/data.
Charts render correctly.
Case queue works.
Client 360 routes work.
Report page works.
Mobile layout is acceptable.
README is updated.
```

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Live dashboard URL	Public URL opens successfully.
GitHub repository	Code and documentation are available.
Updated README	Screenshots, setup instructions, and live link are included.
Deployment notes	Known limitations and deployment details are documented.

16. Phase 12 - RAG Assistant

Objective

Add an AI investigation assistant that can answer questions about a selected client using retrieved evidence. This should be treated as an advanced feature after the dashboard is live.

Important rule: do not put API keys in the React frontend. Use a backend API, Next.js API route, or serverless function for any LLM calls.

Backend Options

```
Next.js API route
FastAPI backend
Serverless function
```

RAG Knowledge Base

```
Client profile
Screening hits
Ownership records
Adverse media articles
Previous cases
Country risk notes
Generated reports
```

Example Client Document

```
Client: ABC Global Holdings
Industry: Logistics
Countries: UAE, Russia, Singapore
Risk Score: 92
Risk Level: High
Risk Drivers: High adverse media, high-risk country exposure, internal alert
Adverse Media: Bribery allegation in infrastructure contract
Recommendation: Enhanced Due Diligence
```

RAG Questions

```
Why is this client high risk?
What are the main risk drivers?
Is this hit likely true or false positive?
Summarize adverse media for this client.
What evidence supports Enhanced Due Diligence?
What should the analyst review next?
```

RAG Answer Format

```
Answer
Evidence
Source references
Confidence
Suggested next step
```

Example RAG Answer

```
ABC Global Holdings is classified as High Risk because it has one high-severity adverse media record, exposur
high-risk jurisdiction, and one previous internal alert.

Evidence:
1. Adverse media article A021 mentions bribery allegations.
2. Country risk table marks Russia as high risk.
3. Previous case C019 was escalated in 2022.

Suggested next step:
Perform Enhanced Due Diligence and send the case for senior review.
```

Tasks

- Create client text documents from structured data.
- Create embeddings for client documents.
- Store embeddings in a vector database such as ChromaDB or FAISS.
- Build retrieval function to fetch relevant client evidence.
- Create backend endpoint for question answering.
- Connect React UI to backend endpoint.
- Show answer with citations or evidence sections.
- Add guardrails for insufficient evidence.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
RAG assistant page	User can ask client-specific questions.
Backend API	Questions are processed securely without exposing API keys.
Grounded answers	Assistant returns evidence-backed responses.
Source references	Important claims show the documents used.
Guardrails	Assistant avoids unsupported claims when evidence is missing.

17. Phase 13 - Testing and Quality Assurance

Objective

Ensure the live app is stable, accurate, and portfolio-ready. Testing should cover data, UI, deployment, and business logic.

Data Testing

```
All clients have risk scores.
All clients have risk levels.
No missing client IDs.
No broken client links.
Risk score calculations are correct.
```

UI Testing

```
All pages load.
Charts display correctly.
Tables filter correctly.
Client profile opens correctly.
Report generation works.
```

Deployment Testing

```
Live URL works.
Routes do not break on refresh.
Static data loads correctly.
No API key is exposed.
No console errors.
```

Business Logic Testing

Test Case	Expected Result
Critical sanctions case	Should be classified as Critical.
False positive case	Should not become High Risk only due to name similarity.
High adverse media case	Should be High or Critical.
PEP case	Should be Medium or High depending on other risk factors.
Offboarded client	Should show correct status and adjusted recommendation.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Testing checklist	Data, UI, deployment, and business logic tests are documented.
Bug fixes	Critical issues are resolved.
Stable live version	App is usable and shareable.

18. Phase 14 - Documentation and Portfolio Packaging

Objective

Prepare the project for MSBA portfolio, GitHub, resume, and interview discussion. The documentation should communicate both technical implementation and business value.

README Structure

```
Project Title
Problem Statement
Business Context
Solution Overview
Tech Stack
Data Sources
System Architecture
Dashboard Features
Risk Scoring Logic
RAG Assistant Explanation
Screenshots
Live Demo Link
How to Run Locally
Limitations
Future Improvements
```

Portfolio Write-Up Structure

```
Problem:
Compliance analysts manually review fragmented KYC/AML data across multiple files and systems.

Solution:
Built a live due diligence risk intelligence dashboard that consolidates data, scores risk, prioritizes cases
generates evidence-backed reports.

Impact:
The platform improves visibility into high-risk clients, reduces manual data lookup, standardizes risk scorin
supports better case prioritization.
```

Resume Bullet Example

> Built a live AI-assisted due diligence risk dashboard using React, Python, Pandas, and rule-based analytics to consolidate synthetic KYC/AML datasets, score client risk, prioritize compliance cases, and generate evidence-backed due diligence summaries.

Deliverables and Exit Criteria

Deliverable	Completion Criteria
Final README	Explains problem, approach, architecture, and usage.
Screenshots	Dashboard, case queue, client 360, and report pages are shown.
Architecture diagram	Shows data pipeline to frontend flow.
Live URL	Included in README and portfolio.
Sample reports	Two or three due diligence report samples are included.
Portfolio summary	Clear story for MSBA and interview discussions.

19. Suggested Timeline

4-Week Aggressive Plan

Week	Focus	Deliverables
Week 1	Data and scoring	Synthetic data, Python pipeline, risk scoring
Week 2	React dashboard	Executive dashboard, case queue
Week 3	Client 360 and reports	Client profile, analytics page, report page
Week 4	Deployment and polish	Live URL, README, testing, portfolio write-up

6-Week Safer Plan

Week	Focus	Deliverables
Week 1	Planning and data	Scope, data dictionary, synthetic CSVs
Week 2	Data pipeline	Cleaning, merging, feature creation
Week 3	Risk engine	Entity matching, risk scoring, JSON exports
Week 4	React dashboard	Dashboard, case queue, charts
Week 5	Client 360 and report	Client details, score explanation, report generation
Week 6	Deployment and RAG MVP	Live deployment, simple RAG assistant, documentation

20. Recommended MVP Cut

To go live quickly, build the core dashboard first and defer advanced features. This approach gives you a public portfolio artifact faster.

Build First

```
Phase 1: Synthetic data
Phase 2: Python processing
Phase 3: Entity matching
Phase 4: Risk scoring
Phase 5: React app setup
Phase 6: Executive dashboard
Phase 7: Case queue
Phase 8: Client 360
Phase 11: Deployment
```

Add Later

```
Risk analytics
PDF report
RAG assistant
Scenario analysis
Authentication
Backend database
```

21. Final Priority Order

Priority	Feature
1	Business problem clarity
2	Data pipeline
3	Risk scoring and explainability
4	Live dashboard
5	Client 360 view
6	Report generation
7	RAG assistant
8	Advanced backend

22. Questions to Finalize Before Development

Product Scope Questions

No.	Question
1	Should the first live version include RAG, or should RAG be Phase 2 after dashboard deployment?
2	Should the dashboard be public without login, or should it include basic authentication?
3	Should the dashboard show only synthetic data, or should users be able to upload their own CSV files later?
4	Should the project be positioned mainly for MSBA admissions, resume, or GitHub portfolio?

Technical Questions

No.	Question
5	Do you want to use Next.js or simple React Vite?
6	Do you prefer TypeScript or JavaScript?
7	Do you want Tailwind CSS, Material UI, or plain CSS?
8	Do you want Recharts, Chart.js, or ECharts?
9	Do you want report download as PDF in MVP, or is copy/download as text enough initially?

Data Questions

No.	Question
10	How many synthetic clients should we start with: 100, 500, or 1,000?
11	Should exact sample rows be defined for all CSV files?
12	Should the data include Indian, US, European, or global corporate names?
13	Should the dashboard include individual clients, corporate clients, or only corporate clients?

AI/RAG Questions

No.	Question
14	Do you want to use OpenAI API for RAG?
15	Should RAG work only for selected clients or across the full knowledge base?
16	Should answers include citations and evidence?
17	Should the RAG assistant be deployed live or shown as a local demo/notebook?

23. Recommended Immediate Next Step

The fastest path to a strong live version is to begin with the synthetic data and processed JSON pipeline. Once the data foundation is ready, the React pages become much easier to build.

```
1. Create GitHub repo
2. Create synthetic CSV data
3. Write Python script to process data
4. Export dashboard JSON
5. Create React app
6. Build executive dashboard
7. Build case queue
8. Build Client 360
9. Deploy on Vercel
```

The first live version should prove the core idea: fragmented due diligence data -> unified client profile -> explainable risk score -> analyst dashboard -> client-level recommendation.